Tracking Motorcycle Performance: An Analysis of Riding Patterns for Fuel Consumption Optimization

Jezreel James M. Hallasgo
College of Computing and Information Technologie
National University- Manila
Manila, Philippines
hallasgojm@students.national-u.edu.ph

Abstract - In this paper, a motorcycle performance tracking approach is presented as a practical way to improve fuel consumption by analyzing riding patterns. Many riders rely only on experience to estimate fuel efficiency, which can be inaccurate and inconsistent. To address this problem, this study analyzes riding behavior using simple tracking methods to identify factors that contribute to fuel inefficiency.
Data were collected over nine weeks (36 trips) using ride tracking methods (speed tracker app) and fuel usage logs to measure variations in fuel efficiency. After cleaning and preprocessing, exploratory data analysis (EDA), Pearson correlation, and ANOVA were applied. Results showed that distance traveled and total travel time were the strongest predictors of fuel consumption, demonstrating a near linear relationship with fuel usage. Using the Ridge regression model, it shows predictive performance ($R^2 = 0.96$), confirming that distance, and total travel time explain most of the fuel consumption changes.

Keywords - Tracking, Fuel Consumption, Riding patterns, Riding behavior

I. Introduction

Motorcycles are widely used due to their affordability, maneuverability, and fuel efficiency, making them a practical mode of transportation for daily commuting and delivery services. In the Philippines, motorcycles play a vital role in personal mobility, particularly in congested urban areas where traffic conditions influence travel efficiency.

This topic is important because rising fuel prices and environmental concerns highlight the need for efficient fuel use. Understanding how riding behavior affects fuel consumption allows riders to adopt habits that promote both economic savings and environmental sustainability.

Previous studies have shown that driving behavior significantly influences fuel consumption. Aggressive acceleration, sudden braking, prolonged idling, and inconsistent speeds increase fuel usage, while smooth acceleration and steady speeds improve fuel efficiency. A study demonstrated that driver behavior can be analyzed and classified to predict fuel consumption, highlighting the strong relationship between driving patterns and fuel efficiency.

Despite these findings, limited research focuses specifically on motorcycles operating in real-world urban traffic conditions. Motorcycle riders frequently encounter stop-and-go traffic, varying road environments, and prolonged

idling, which may influence fuel consumption differently from larger vehicles. This project addresses this gap by analyzing actual riding patterns and their relationship to fuel efficiency in everyday riding conditions.

The primary goal of this study is to analyze motorcycle riding patterns and determine how they influence fuel consumption. By tracking performance indicators such as speed, acceleration behavior, idling time, and travel distance, the study aims to identify behaviors that contribute to fuel inefficiency and recommend strategies for optimizing fuel use.

Specifically, this study seeks to answer the following research questions:

1.) What riding patterns are commonly observed among motorcycle users?
2.) How do speed, acceleration, motor load, and traffic affect fuel consumption?
3.) Is there a significant relationship between riding behavior and fuel efficiency?
4.) What riding practices can be recommended to optimize fuel consumption?

The findings of this research may help motorcycle riders improve fuel efficiency, reduce operating costs, and promote environmentally responsible riding practices.

II. Literature reviews

A. Riding Behavior and Fuel Consumption

Rider behavior has a significant impact on motorcycle fuel consumption. Studies have shown that aggressive acceleration, sudden braking, frequent idling, and inconsistent speed patterns increase engine workload and, consequently, fuel usage [1]. Conversely, maintaining smooth acceleration, steady speeds, and controlled braking improves fuel efficiency while reducing emissions [2]. These findings highlight the importance of rider behavior in optimizing fuel consumption, especially in urban areas where stop-and-go traffic and variable road conditions are common.

For motorcycles, which have smaller engines compared to cars, these effects are even more pronounced. Unlike cars, motorcycles are more sensitive to acceleration changes and load variations, making rider behavior a critical factor in real-world fuel efficiency. [2] Eco-driving practices, such as consistent speed maintenance and minimizing idle time, have been shown to improve fuel efficiency in both controlled experiments and real-world urban settings.

B. Idle Time and Traffic Interaction

Idle time directly affects fuel consumption and emissions. Stop-and-go traffic, traffic lights, and congestion increase idle periods, reducing fuel efficiency even when travel distance is the same. [3] found that trips with higher idling times consumed significantly more fuel, highlighting the impact of traffic interactions on motorcycle efficiency.

Reducing idle time through smoother riding and awareness of traffic flow can improve fuel efficiency. Eco-driving studies for motorcycles indicate that minimizing unnecessary idling, combined with consistent speed maintenance, leads to measurable fuel savings [2]. This shows that managing idle time is a practical and effective strategy for optimizing fuel consumption in urban riding conditions.

C. Vehicle Load and Fuel Consumption

Vehicle load, including passengers or cargo, affects motorcycle fuel efficiency by increasing engine workload. Heavier loads require more power for acceleration and climbing, which leads to higher fuel consumption. Studies show that even a single passenger can reduce fuel efficiency slightly, though the impact is more pronounced during frequent stop-and-go riding [1].

For motorcycles, which have smaller engines than cars, extra weight amplifies the effect of acceleration and braking, making load management an important factor for optimizing fuel use. Awareness of load and minimizing unnecessary weight can help riders improve fuel efficiency during daily urban commutes.

D. Real-World Driving Conditions

A 2023 study examining motorcycle fuel use in Vietnam [1] confirmed that actual driving characteristics (including modes of acceleration and speed variations) directly influence fuel consumption, indicating that the complexities of real urban traffic must be considered when analyzing fuel patterns.

These studies collectively underline that fuel consumption in motorcycles is not determined solely by mechanical specifications but by how riding behavior interacts with traffic conditions, speed profiles, and environmental contexts.

E. Research Gap and Relevance to the Study

Research shows that driving behavior, speed, traffic conditions, and vehicle load significantly affect fuel consumption. Most studies using real-world data, sensors, statistical analysis, or simulations find that aggressive acceleration, inconsistent speeds, congestion, and extra load increase fuel use, while smooth driving improves efficiency.

However, most studies focus on cars or controlled environments, with limited attention to motorcycles in real-world urban traffic. Motorcycles differ in engine size, weight, aerodynamics, and rider control, which may affect fuel consumption differently. This study examines motorcycle riding patterns including speed, travel time, riding quality, and backride passengers using accessible tracking methods to offer practical fuel-efficiency recommendations.

III. Methodology

This section describes the procedures used to collect, process, and analyze the trip performance dataset. This outlines the operational definitions of each variable, the data cleaning, and preprocessing steps performed, and the statistical methods applied to evaluate relationships and predictive performance.

A. Participants

The study involved a single participant, the researcher, who conducted all recorded trips. The participant is 22 years old, undergraduate student in National University- Manila. The motorcycle is primarily used as a daily mode of transportation for travel, ensuring that the recorded trips reflect realistic riding conditions rather than controlled experimental rides. No sensitive personal information was collected.

B. Data Collection Methods

Data were collected over a nine week period, covering a total of 36 motorcycle trips. Each trip was recorded immediately after completion to ensure accuracy.

Data Variable	Type (Quant/Qual)	Unit/Scale	Frequency of Collection	Tool/App
Travel Time	Quantitative	Minutes	Per Trip	tracker app
Fuel Consumption	Quantitative	liters	Per Trip	full tank method
Max speed	Quantitative	kph	Per Trip	tracker app
Average Speed	Quantitative	kph	Per Trip	tracker app
Distance	Quantitative	Kilometer	Per Trip	Tracker app
Stopped Time	Quantitative	seconds	Per Trip	Tracker app
Max Alt	Quantitative	Meter	Per Trip	Tracker app
Traffic Level	Qualitative	1-4	Per Trip	Observation
Ride Quality	Qualitative	Slow, Normal, Aggressive	Per Trip	Observation
Backride	Qualitative	0 or 1	Per trip	Observation

C. Operational Definitions

Each variable was defined as follows:

1. Travel Time - Total moving time duration during the trip.
2. Fuel Consumption - Total liters of fuel used for each trip, recorded manually.
3. Max Speed - Highest recorded speed during the trip.
4. Average Speed - Mean speed during the trip.

5. Distance - Total kilometers traveled during the trip.
6. Stopped Time - Total idle time duration during the trip.
7. Max Alt - Highest recorded Altitude during the trip.
8. Traffic Level - Subjective classification of traffic condition(1 - 4).
9. Ride Quality - Self classified riding behavior during the trip (slow, normal, aggressive).
10. Backride - Binary variable indicating the presence of a passenger(0 or 1).
11. Fuel Efficiency - Computed as Distance divided by Fuel Consumption.

D. Data Cleaning and Preprocessing

Before analysis, the dataset underwent cleaning and preprocessing. Text based units (e.g., “kph”, “km”, “L”, “mins”, “seconds”, “m”) were removed.

Missing values were checked, and one outlier was removed due to inaccurate data recorded and affecting the consistency of the result significantly. One variable was added (Efficiency), to ensure the data consistency across all trips.

Categorical variables such as Ride Quality, Traffic Level, and Backride were encoded using one-hot encoding for regression analysis. Numerical variables were standardized using Standard Scaler to ensure comparability across predictors.

E. Statistical Analysis

1. Under exploratory data analysis (EDA), data analysis was conducted to identify patterns and relationships among variables. Scatterplot and Boxplot were used for visualizing the data. Scatterplots were used to visualize the regression result of distance vs fuel consumption, and to identify the relation of fuel consumption (Target Variable) across all numerical variables. Boxplots were used to identify the relation of target variable across all categorical variables such as Ride Quality, Traffic Level, and Backride. A correlation heatmap was generated to examine Pearson correlation coefficients among numerical variables.

2. Pearson correlation analysis was performed to measure the linear relationships between continuous variables and fuel consumption.

Analysis of Variance (ANOVA) was conducted to evaluate whether categorical variables (Ride Quality, Traffic Level, Backride) produced statistically significant differences in fuel consumption.

3 I used the **T-test** to compare **fuel efficiency** between **backride groups** (0 vs 1).

Initially, the **Backride** feature may have had multiple categories or continuous values. To make the analysis more clearer and to apply the T-test effectively, I **grouped the backride values** into two categories:

- **No Backride** (represented as 0)
- **With Backride** (represented as 1)

This grouping simplifies the analysis by creating two distinct groups, making the **comparison between fuel efficiency** in these groups more meaningful.

4. To evaluate the Ridge regression model performance. Model performance was evaluated using:

- Coefficient of Determination (R^2)
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Residual analysis and actual versus predicted plots were examined to assess model accuracy and assumptions.

IV. Results

This section presents the outcomes of the statistical analysis conducted on the 35 recorded Trips(1 outlier recorded data removed). The findings are reported objectively using descriptive statistics, visual summaries, correlations, regression outputs, and hypothesis-testing results.

A. Overview of the Dataset

The dataset consists of 35 ride observations with 11 original variables and one engineered feature (efficiency).

Key Variables:

- Distance (km)
- Fuel Consumption (L)
- Average Speed (kph)
- Max Speed (kph)
- Travel Time (minutes)
- Stopped Time (seconds)
- Ride Quality (Slow, Normal, Aggressive)
- Traffic Level (1–4)
- Backride condition (binary grouped)

No missing values were found in the dataset.

	Variable	Mean	Median	SD	Min	Max
0	Fuel Consumption	1.98	1.81	0.85	0.52	3.64
1	Average Speed	21.38	20.60	3.15	16.30	29.40
2	Efficiency	39.45	38.68	3.94	32.24	47.69
3	Stopped Time	1389.83	1500.00	636.37	300.00	2686.00
4	Distance	77.74	69.40	34.36	24.80	155.70

Fig.1. Descriptive statistics of key numerical variables

The figure shows that, on average, trips last for about **77 km**, with fuel consumption averaging 1.98 **liters per trip**. So on average, it cost me more than 100 pesos per trip, that goes 39.45 km per liters

	Date	Max speed	Average Speed	Distance	Travel Time	Stopped Time	Traffic Level	Ride Quality	Backride	Fuel Consumption	Max Alt	Efficiency
0	19-Nov-25	88.3	20.7	43.6	117.0	387.0	2	Aggressive	0.50	1.20	76.0	36.333333
1	20-Nov-25	90.3	21.8	39.2	101.0	528.0	2	Aggressive	0.50	1.05	77.0	37.333333
2	21-Nov-25	54.3	18.6	44.3	135.0	1130.0	3	Slow	0.00	0.94	41.0	47.127660
3	22-Nov-25	89.8	29.4	80.8	160.0	745.0	1	Aggressive	1.00	1.96	75.0	41.224490
4	24-Nov-25	56.0	17.0	43.4	144.0	934.0	2	Slow	0.50	1.12	79.0	38.750000
5	26-Nov-25	88.5	19.8	91.4	252.0	2280.0	3	Aggressive	0.50	2.34	77.0	39.059829
6	28-Nov-25	90.3	20.1	49.9	135.0	1500.0	3	Aggressive	0.50	1.29	77.0	38.682171
7	29-Nov-25	87.2	20.0	69.4	197.0	1130.0	2	Aggressive	1.00	1.95	78.0	35.589744
8	2-Dec-25	76.7	23.1	45.0	109.0	1200.0	3	Normal	0.00	1.00	45.0	45.000000
9	3-Dec-25	89.7	24.1	45.0	104.0	969.0	2	Aggressive	0.33	1.19	87.0	37.815126

Fig. 2. shows the dataset sample.

B. Distribution of Key Variables and Their Relationships

This section explores the distribution of key variables in the dataset and examines the relationships between different variables.

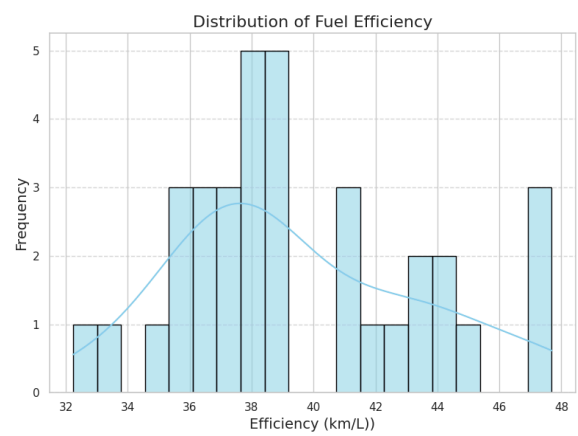

Fig. 3. Distribution of Fuel Efficiency. Histogram showing the frequency distribution of Efficiency across all trips. The overlaid curve represents the density estimation.

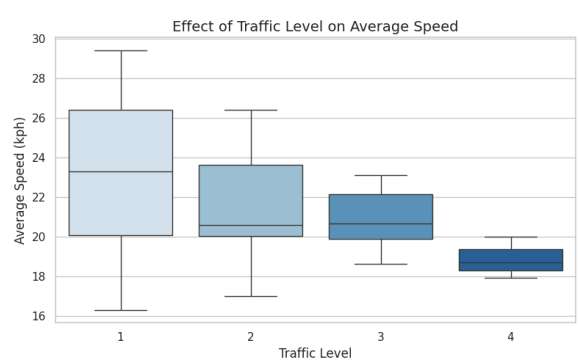

Fig. 4. Traffic Level vs Average Speed. Boxplot showing the distribution of average speed across different traffic levels.

Fig. 5. Fuel Efficiency Across Different Ride Qualities. Boxplot showing the distribution of fuel efficiency for slow, normal, and aggressive driving styles. Green represents slow, yellow normal, and red aggressive driving.

C. *Correlation Matrix and Exploratory Findings*

This section presents the correlation analysis and explores the relationships between fuel consumption and key variables in the dataset.

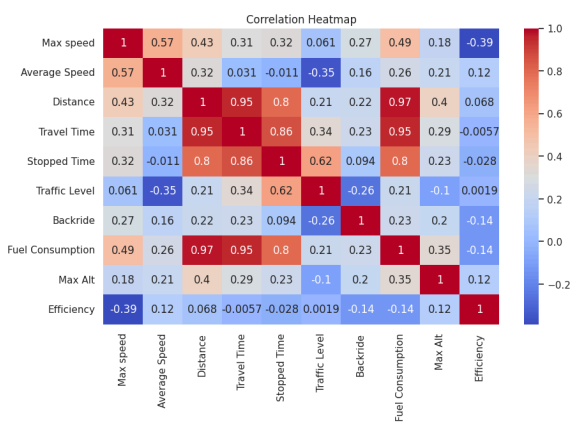

Fig. 6. Correlation heatmap of numeric variables in the dataset, showing the strength and direction of linear relationships. Positive correlations are shown in red, negative correlations in blue.

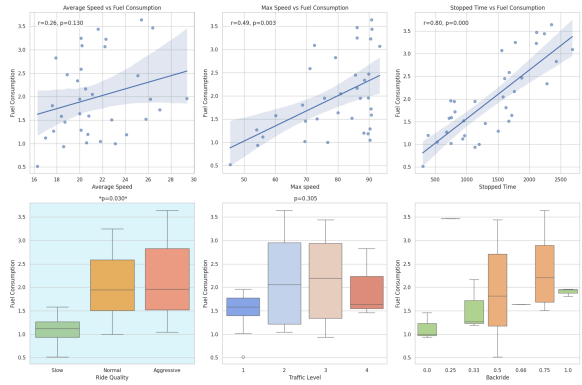

Fig. 7. Relationships between fuel consumption and key continuous and categorical variables. Panels (a–c) show scatterplots with regression lines for continuous variables (*Average Speed*, *Max Speed*, and *Stopped Time*) including Pearson correlation coefficients (r) and p-values. Panels (d–f) show boxplots for categorical variables (*Ride Quality*, *Traffic Level*, and *Backride*), with light shading indicating statistically significant differences by one-way ANOVA ($p < 0.05$).

D. *Regression Analysis: Distance vs Fuel Consumption*

This section explores the relationship between trip distance and fuel consumption through a linear regression analysis.

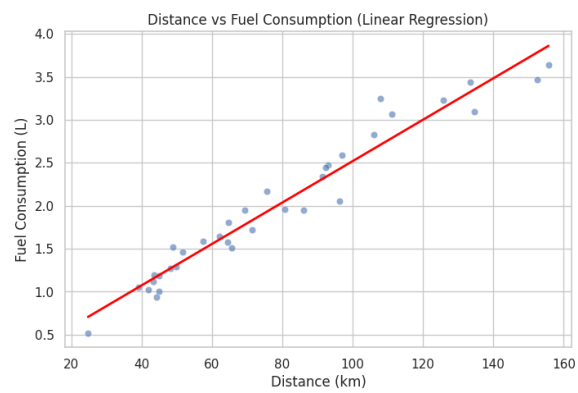

Fig. 8. Distance vs Fuel Consumption. Scatter plot showing fuel consumption across different trip distances. The red line represents a linear regression fit, indicating the trend of increasing fuel consumption with distance.

E. *Impact of Backride on Fuel Efficiency*

This section examines the influence of the Backride condition on fuel efficiency across different visualizations. The analysis explores the differences in fuel efficiency between trips with and without a backride.

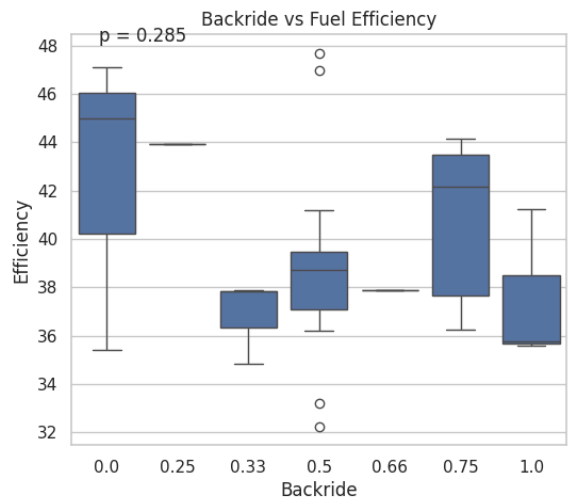

Fig. 9. Backride vs Fuel Efficiency. Boxplot showing fuel efficiency for trips with and without a backride. The distributions are highly variable and do not show a clear separation between groups, indicating that Backride is not a strong predictor of efficiency; thus, it was excluded as a feature in the model.

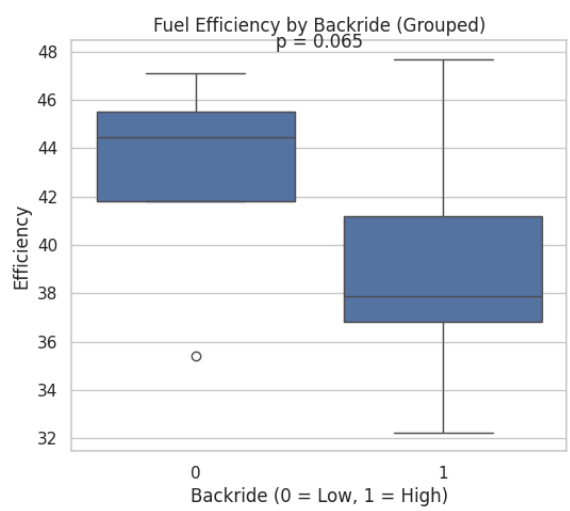

Fig. 10. Fuel Efficiency by Backride (Grouped). Boxplot comparing fuel efficiency for trips without backride (0) and with backride (1). Grouping clarifies the separation, and since the difference is close to significance ($p \approx 0.05$), Backride was retained as a feature in the predictive model.

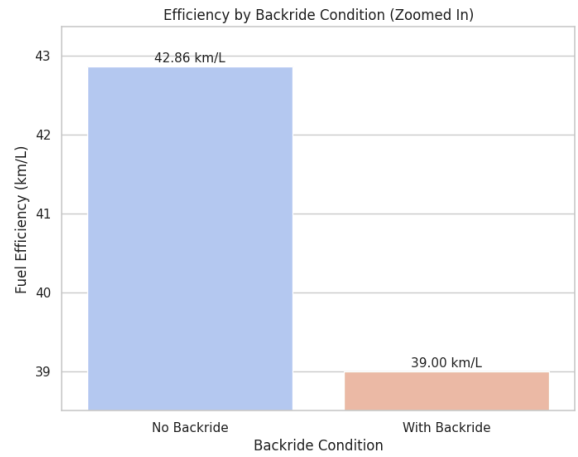

Fig. 11. Fuel Efficiency by Backride Condition. Bar plot showing the mean fuel efficiency for trips without backride ("No Backride") and with backride ("With Backride"). Values are annotated above each bar. The plot highlights that trips with backride tend to have slightly lower efficiency, supporting its inclusion as a predictive feature.

F. *Model Evaluation: Actual vs Predicted Fuel Consumption*

This section evaluates the performance of the regression model by comparing predicted fuel consumption values with the actual observed values.

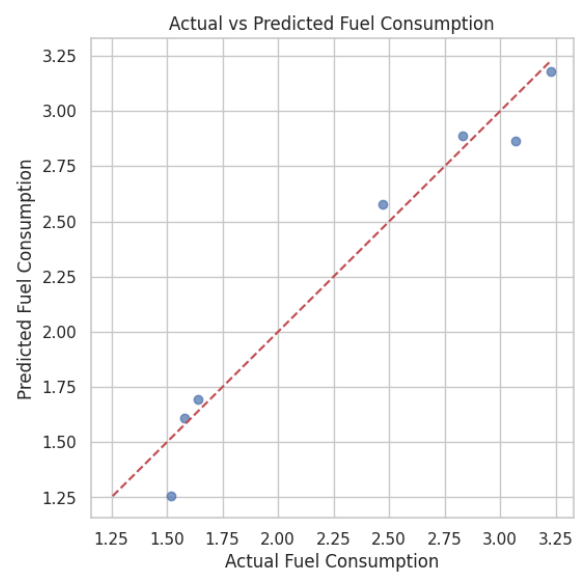

Fig. 12. Actual vs Predicted Fuel Consumption. Scatter plot comparing the predicted fuel consumption from the regression model against the actual values. The red dashed line represents perfect predictions ($y = x$). Points close to this line indicate

accurate predictions, showing that the model reasonably captures the variation in fuel consumption.

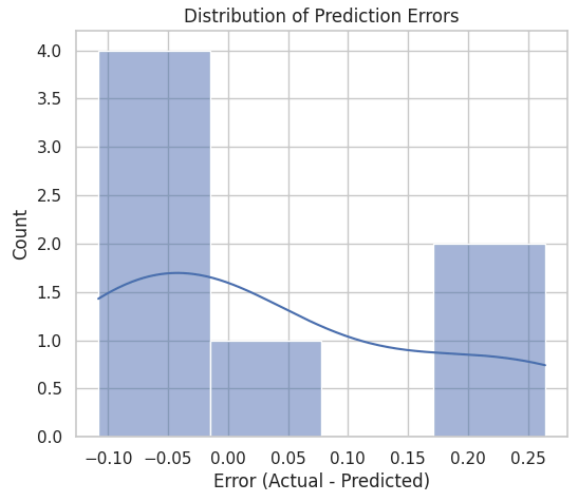

Fig. 13. Distribution of Prediction Errors. Histogram of the residuals (actual minus predicted fuel consumption) with a KDE overlay. The distribution is approximately centered around zero, indicating that the model has no strong systematic bias, and most prediction errors are small.

G. Impact of Stopped Time on Fuel Efficiency

This analysis examines how fuel efficiency varies between low and high stopped time groups (split at the median)."

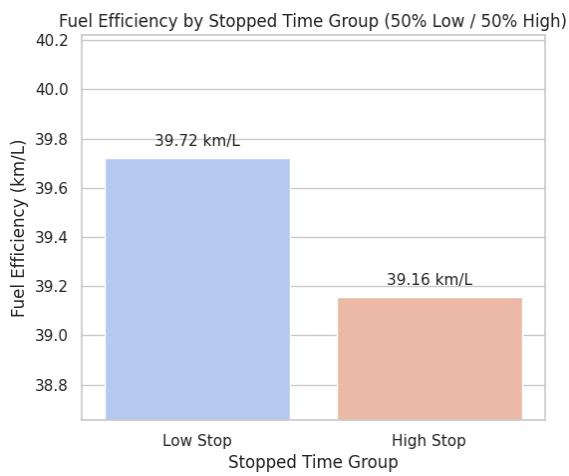

Fig. 14. Fuel Efficiency by Stopped Time Group. Bar plot showing mean fuel efficiency for trips with low vs. high stopped time (split at the median). Trips in the Low Stop group exhibit higher fuel efficiency than those in the High Stop group, indicating that longer stopped durations negatively impact efficiency.

H. Impact of Ride Quality and Backride on Fuel Consumption

This section examines the effects of Ride Quality and Backride on fuel consumption, based on model coefficients.

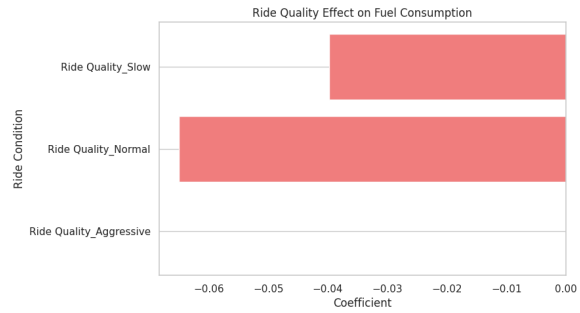

Fig. 15. Effect of Ride Quality on Fuel Consumption. Horizontal bar plot showing the model coefficients for different Ride Quality categories. Positive coefficients indicate higher fuel consumption relative to the baseline category (Aggressive), while negative coefficients indicate lower consumption. This highlights how driving behavior impacts fuel efficiency.

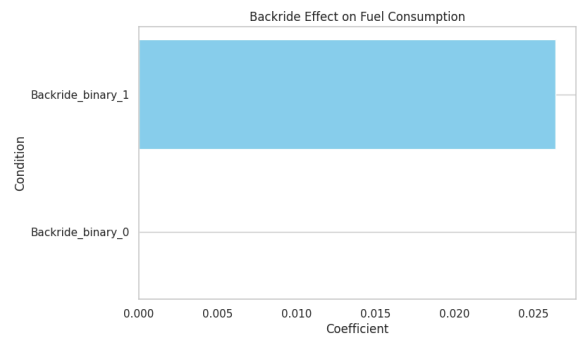

Fig. 16. Backride Effect on Fuel Consumption. Bar plot of model coefficients showing how the presence of a backride (1) affects fuel consumption relative to no backride (0).

The analysis indicates that while the difference in fuel efficiency between trips with and without a backride is close to statistical significance ($p \approx 0.05$), the effect of Backride on fuel efficiency is worth considering in the model.

I. Comparison result of riding pattern

The analysis of fuel consumption across different driving patterns and conditions reveals several key insights.

Driving Pattern Comparison:

Aggressive driving resulted in the highest fuel consumption at **1.356 L** per trip. In contrast, **slow driving** showed a **14.5% reduction** in fuel consumption (1.159 L), while **normal driving** yielded a **16.2% reduction** (1.137 L). These findings emphasize the significant role driving style plays in fuel efficiency, with smoother driving leading to better fuel economy.

Impact of Backride:

The presence of a **backride** had a minimal impact on fuel consumption. Trips with a backride consumed **1.356 L**, identical to aggressive driving, while trips without a backride showed a slight reduction to **1.329 L**, a **2% improvement**. This suggests that the backride condition has a negligible effect on fuel efficiency.

Optimal vs. Worst Riding Patterns:

When adopting an **optimal riding pattern**, fuel consumption decreased to **1.105 L**, representing an **18.5% improvement** over aggressive driving. Even doubling the distance with the optimal pattern led to a moderate increase to **2.258 L**, but it still remained more efficient than aggressive driving. Conversely, the **worst riding pattern** resulted in the highest fuel consumption (1.356 L), and when the distance was doubled, fuel consumption increased to **2.561 L**, showing **11.83%** increase over the optimal riding pattern doubled distance demonstrating that poor driving habits make fuel inefficiency worse.

V. Discussion

A. Interpretation of Results

The results of the study reveal several key insights into the factors influencing fuel consumption across different driving patterns. The data indicates a clear relationship between **driving style** and fuel efficiency, with **aggressive driving** showing the highest fuel consumption at **1.356 L** per trip. In contrast, **slower driving styles** led to significant reductions in fuel consumption, with **slow driving** resulting in a **14.5% reduction** (1.159 L) and **normal driving** showing an even greater improvement of **16.2%** (1.137 L). These findings suggest that smoother, more controlled driving habits can lead to better fuel economy, supporting prior research which shows that gradual acceleration and deceleration are key factors in reducing fuel consumption.

The impact of the **backride condition** was minimal in this analysis. Trips with a backride consumed **1.356 L**, identical to aggressive driving, while trips without a backride showed only a small reduction to **1.329 L**, representing a **2% improvement**. This suggests that while carrying extra weight may intuitively lead to higher fuel consumption, it has little to no substantial effect on overall fuel economy, at least in comparison to the influence of driving style.

Furthermore, the comparison between **optimal and worst riding patterns** reveals striking differences in fuel consumption. The **optimal riding pattern**, characterized by smoother driving, resulted in a fuel consumption of **1.105 L**, an **18.5% improvement** over aggressive driving. Even when the trip distance was **doubled** while maintaining the optimal pattern, the fuel consumption only increased to **2.258 L**, which was still more efficient than aggressive driving. In contrast, the **worst riding pattern**, which mirrored the high fuel consumption of

aggressive driving, demonstrated that poor driving habits make fuel inefficiency worse. When the distance was **doubled**, fuel consumption increased to **2.561 L**, representing an **11.83% increase** compared to the optimal pattern with doubled distance. This reinforces the idea that driving behavior plays a pivotal role in fuel consumption, and inefficient driving habits can have a significant negative impact, especially over longer trips.

These findings are consistent with prior literature that highlights the importance of driving behavior in determining fuel efficiency, with smoother, more controlled driving being linked to improved fuel economy . However, the analysis also underscores that **backride condition**, despite its theoretical impact, does not substantially alter fuel consumption, indicating that driving habits outweigh other variables such as weight carried during the trip.

In conclusion, the results highlight the importance of adopting an optimal riding pattern for improved fuel efficiency, with the potential for **18.5% reduction** in fuel consumption compared to aggressive driving. Although **backride condition** has minimal effect, maintaining smooth driving habits can lead to significant fuel savings, especially on longer trips. These findings emphasize the crucial role of driving behavior in achieving fuel efficiency, offering valuable insights for individuals and operators seeking to reduce fuel consumption.

B. Comparison to Related Work

The findings of this study support the established view that riding behavior plays a major role in fuel consumption. As discussed in the previous studies, aggressive acceleration, sudden braking, and inconsistent speed patterns increase fuel usage [1]. The present results confirm that aggressive riding produced the highest fuel consumption, while smoother riding styles improved fuel efficiency. This reinforces the importance of controlled acceleration and steady speed maintenance highlighted in eco-driving research [2].

The study also supports prior research on idle time and traffic interaction. Previous findings indicate that stop-and-go traffic and prolonged idling increase fuel consumption even when travel distance remains constant [3]. Similarly, trips with longer stopped time in this study showed lower fuel efficiency, demonstrating the negative effect of congestion and traffic signals in real-world urban riding conditions.

In terms of vehicle load, earlier studies suggest that additional weight increases engine demand, particularly during acceleration and frequent stops [1]. Although only a minimal difference was observed when a passenger was present, the slight reduction in efficiency aligns with literature indicating that added load can influence fuel consumption.

Consistent with research emphasizing real-world driving conditions, the findings confirm that fuel consumption is shaped not only by mechanical factors but also by how riders interact with traffic flow, speed variations, and urban environments [1]. These results reinforce the importance of examining motorcycle fuel efficiency within actual riding contexts rather than controlled settings.

C. Limitations

Several limitations should be considered when interpreting the results of this study:

- **Small Dataset -** The dataset consisted of only 35 recorded trips, which limits the ability to generalize the findings across different driving conditions, traffic level, time periods.
- **Single Participants -** Data was collected by a single researcher, the result may reflect the specific driving

behavior towards and cannot be represented to a broader population.

- **Short data collection window -** Approximately two months may not capture long-term patterns or seasonal effects.
- **Self-Reported Variables** – Variables such as **Ride Quality** and **Traffic Level** were subjectively classified, which could introduce bias in assessing driving behavior, and trip condition and affect the result of the data.

D. Recommendations and Future Work

Practical Recommendations

- **Steady Speed** – Maintain a consistent pace to reduce unnecessary fuel use.
- **Smooth Acceleration** – Increase speed gradually to avoid excessive engine workload.
- **Gentle Braking** – Slow down smoothly to conserve momentum and fuel.
- **Traffic Awareness** – Anticipate traffic flow to minimize stopping and idling.
- **Reduced Idling** – Turn off the engine during long stops when safe.
- **Light Load** – Avoid carrying unnecessary weight to lessen engine strain.

Recommendations for Future Research

Future studies may improve upon this research by:

- Including multiple riders and motorcycle types for broader applicability.
- Using sensor-based measurements for acceleration and throttle behavior.
- Examining environmental variables such as road gradient and weather conditions.
- Expanding the dataset to include longer monitoring periods and varied routes.
- Comparing fuel efficiency between different engine displacements or fuel types.

VI. Conclusion

Key Findings:

This study revealed several important factors influencing motorcycle fuel consumption.

•**Distance & Travel Time -** These were the strongest predictors of fuel consumption, showing a near-linear relation with fuel use.

•**Riding Behavior -** Aggressive driving resulted in the highest fuel consumption, while smooth and steady riding improved fuel efficiency.

•**Idling Time -** Longer idle periods reduced fuel efficiency, highlighting the impact of stop-and-go traffic.

•**Vehicle Load -** Carrying a passenger slightly reduces fuel efficiency, but its impact is minimal compared to riding behavior.

•**Model Accuracy -** The ridge regression model ($R^2 = 0.96$) confirmed that distance and travel time explain most variations in the fuel consumption.

Personal knowledge acquired

This study highlights the significant impact of riding behavior and condition on fuel efficiency. I learned how smooth flow of riding directly influences fuel efficiency. I discovered that maintaining a steady speed by doing smooth throttling, avoiding aggressive acceleration, and sudden braking, and minimizing idling can all lead to reduction of fuel consumption. In addition, the presence of a backride also had a little impact on fuel efficiency, emphasizing that the rider's behavior, and trip conditions plays a much larger role.

This has enhanced my understanding of how small adjustments to my riding style can lead to better fuel efficiency, lower costs, and lower the environmental impact caused by

excessive fuel consumption, less fuel = less pollution.

Final Conclusion

This study demonstrated that motorcycle fuel consumption depends not only on distance but also on riding patterns. By observing my own trips, I learned that aggressive acceleration, often stops, and long idling increase fuel use, while smooth riding and constant speeds improve efficiency. These results made me more aware of my own riding habits and how small changes like anticipating traffic, riding steadily, and minimizing unnecessary stops can lower fuel use, save money, and reduce environmental impact. Overall, tracking performance and adjusting my behavior provides a practical way to ride, more efficiently and responsibly in daily life.

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